Description

**Context**

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. Online food delivery service is a great option for them. It provides them with good food from their favorite restaurants. A food aggregator company FoodHub offers access to multiple restaurants through a single smartphone app.

The app allows restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirms the pick-up in the app and travels to the customer's location to deliver the food. The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. The food aggregator earns money by collecting a fixed margin on the delivery order from the restaurants.

**Objective**

The food aggregator company has stored the data of the different orders made by the registered customers in their online portal. They want to analyze the data to get a fair idea about the demand of different restaurants which will help them in enhancing their customer experience. Suppose you are hired as a Data Scientist in this company and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company improve its business.

**Data Description**

The data includes various information related to a food order. A detailed data dictionary is provided below.

Data Dictionary

- order_id: Unique ID of the order
- customer_id: ID of the customer who ordered the food
- restaurant_name: Name of the restaurant
- cuisine_type: Cuisine ordered by the customer
- cost_of_the_order: Cost of the order
- day_of_the_week: Indicates whether the order is placed on a weekday or weekend (The weekday is from Monday to Friday and the weekend is Saturday and Sunday)
- rating: Rating given by the customer out of 5
- food_preparation_time: Time (in minutes) taken by the restaurant to prepare the food. This is calculated by taking the difference between the timestamps of the restaurant's order confirmation and the delivery person's pick-up confirmation.

- delivery_time: Time (in minutes) taken by the delivery person to deliver the food package. This is calculated by taking the difference between the timestamps of the delivery person's pick-up confirmation and drop-off information

**Submission Guidelines**

1. There are two ways to work on this project:

**i. Full-code way:** The full code way is to write the solution code from scratch and only submit a final Python notebook with all the insights and observations.

**ii. Low-code way**. The low-code way is to use an existing solution notebook template to build the solution and then submit a business presentation with insights and recommendations.

The primary purpose of providing these two options is to allow learners to opt for the approach that aligns with their learning aspirations and outcomes. The below table elaborates on these two options.

| Submission type | Who should choose | What is the same across the two | What is different between the two | Final submission file [IMP] | Submission Format |
|---|---|---|---|---|---|
| Full-code | Learners who aspire to be in hands-on coding roles in the future focussed on building solution codes from scratch | Perform exploratory data analysis to identify insights and recommendations for the problem | Focus on code writing: 10 - 20% grading on the quality of the final code submitted | Solution Python notebook from the full-code template submitted in .html format | .html |
| Low-code | Learners who aspire to be in managerial roles in the future - focus on solution review, interpretation, recommendations, and communicating with business | Focus on business presentation: 10 - 20% grading on the quality of the final business presentation submitted | Business presentation in .pdf format with problem definition, insights, and recommendations | .pdf | |

Please follow the below steps to complete the assessment.Kindly note that if you submit a presentation, ONLY the presentation will be evaluated. Please make sure that all the sections mentioned in the rubric have been covered in your submission.

**i. Full-code version**

- Download the full-code version of the learner notebook.
- Follow the instructions provided in the notebook to complete the project.
- Write down insights and recommendations for the business problems in the comments.
- Submit only thesolution notebook prepared from the learner notebook [format: .html]
- Use the following link to convert **ipynb into html - Html Converter**

**ii. Low-code version**

- Download the low-code version of the learner notebook.
- Follow the instructions provided in the notebook to complete the project.
- Prepare a business presentation with insights and recommendations for the business problem.
- Submit only thepresentation [format: .pdf]

2. Any assignment found copied/plagiarized with other submissions will not be graded and awarded zero marks.

3. Please ensure timely submission as any submission post-deadline will not be accepted for evaluation.

4. Submission will not be evaluated if

- it is submitted post-deadline, or
- if more than 1 file is submitted.

**Best Practices for Full-code Submissions**

- The final Python notebook should be well-documented, with inline comments explaining the functionality of code and markdown cells containing comments on the observations and insights.
- The notebook should be run from start to finish sequentially before submission.
- It is important to remove all warnings and errors before submission.
- The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb).
- Please refer to the FAQ page for common project-related queries.

**Best Practices for Low-code Submissions**

- The presentation should be made keeping in mind that the audience will be the Data Science lead of a company.
- The key points in the presentation should be the following:
    - Business Overview of the problem and solution approach
    - Key findings and insights that can drive business decisions
    - Business recommendations
    - Focus on explaining the key takeaways in an easy-to-understand manner.
    - The inclusion of the potential benefits of implementing the solution will give you the edge.
- Copying and pasting from the notebook is not a good idea, and it is better to avoid showing codes unless they are the focal point of your presentation.
- The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file.
- Please refer to the FAQ page for common project-related queries.

**Power Ahead!**

Rubric Evaluated

Criteria

Understanding the structure of the data

- Overview of the dataset shape, datatypes
- Statistical summary and check for missing values
- Answer all the key questions asked in this section

Comments

Areas performed well:
1. Provided an overview of the dataset's shape and data types.
2. Conducted a statistical summary.
3. Addressed all the key questions in this section.

Points

5/5

Criteria

Univariate Data Analysis

- Explore all the variables and provide observations on the distributions of all the relevant variables in the dataset
- Answer all the key questions asked in this section

Comments

Areas performed well:
1. Excellent work on conducting a univariate analysis, including the creation of histograms and box plots for key variables
2. Successfully identified the top 5 restaurants with the highest number of orders.
3. Effectively determined the most popular cuisine on weekends.
4. Accurately calculated the percentage of orders costing more than $20.
5. Successfully computed the mean order delivery time based on the dataset.
6. Identified the IDs of the top 3 frequent customers along with their order details.

Points

15/15

Criteria

Multivariate Data Analysis

- Perform bivariate/multivariate analysis to explore relationships between the important variables in the dataset
- Answer all the key questions asked in this section

Comments

Areas performed well:

• Excellent work on conducting bivariate and multivariate analyses, including plotting and examining relationships between key variables

- Correctly identified the restaurants that qualify for the promotional offer.

Successfully determined the total revenue generated from all orders in the dataset.

- Impressive calculation of the percentage of orders that took more than 60 minutes to be delivered from the time they were placed.

- Fantastic job in finding the average delivery time for both weekdays and weekends.

Points

20/20

Criteria

Quality & Use of visualisations

- Use proper visualizations for the analysis and provide observations on the plots

Comments

Areas performed well:
1. Excellent work on creating graphs.
2. Selected the appropriate scale for the axes and apt usage of observations.

Points

6/6

Criteria

Conclusion and Recommendations

- Conclude with the key insights/observations

Comments

Good job on providing more than three business conclusions and recommendations derived from the conducted analysis, as this aids business in comprehending the most effective practices and areas requiring enhancement

Points

6/6

Criteria

Presentation/Notebook - Overall Quality

- Structure and flow
- Crispness
- Visual appeal
- All key insights and recommendations covered?

OR

- Structure and flow
- Well commented code
- All key insights and recommendations covered?

Comments

Overall its good, comments are neat and clear in most of the places, Visual representation is also good.

Points

8/8